MILITARY STANDARD

TRUCKLOADING
GRAIN PROPELLANT MK 80 MOD 0
IN PALLET CRATE MK 107 MOD 0
DOMESTIC UNIT LOAD

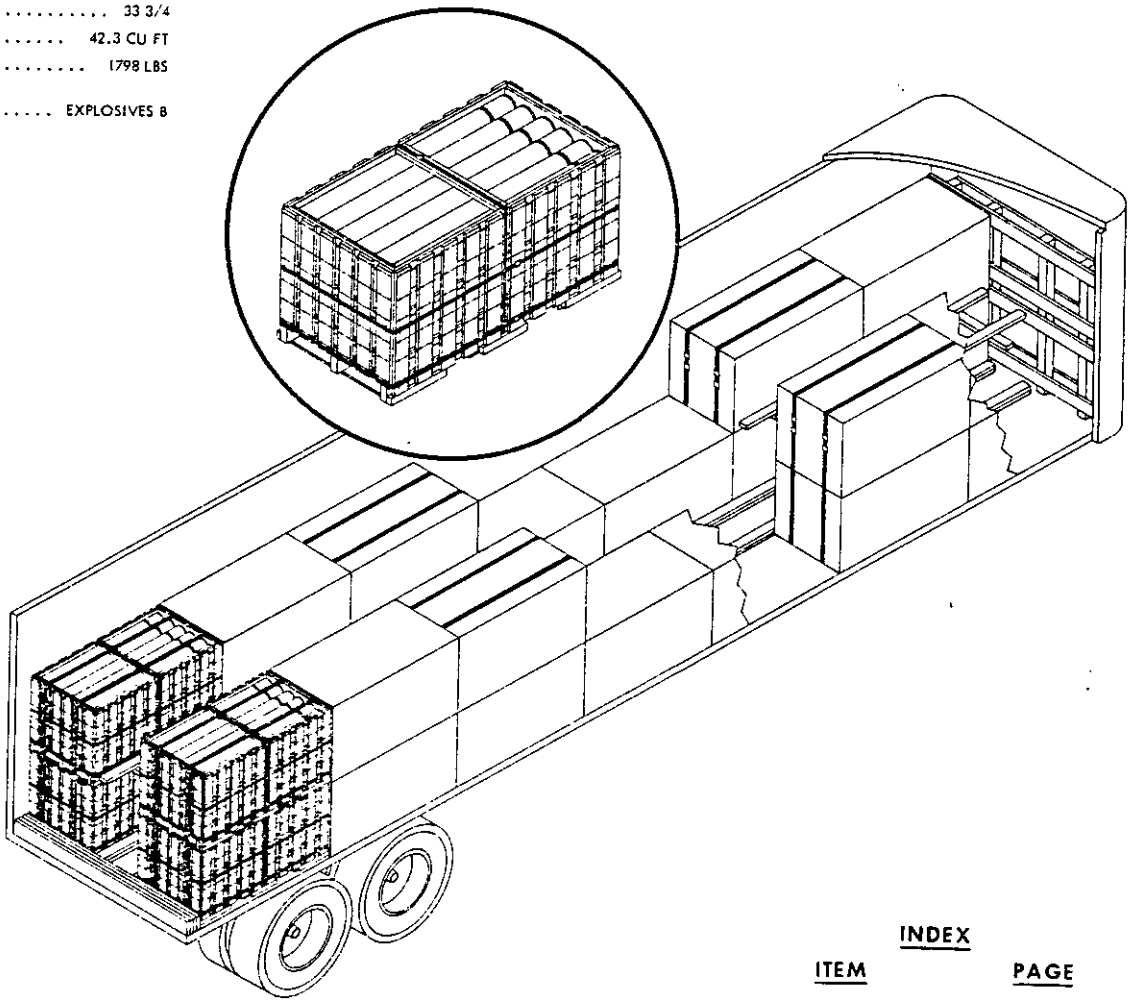
MIL-STD-1320-116
(NAVY)

8 FEBRUARY 1973

UNIT LOAD DATA

UNIT LOAD DRAWING WR-53/845

LENGTH	65 1/4
WIDTH	33 1/4
HEIGHT	33 3/4
CUBE	42.3 CU FT
WEIGHT	1798 LBS
DOT HAZARD CLASSIFICATION	EXPLOSIVES B



NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINER LOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236.

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AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M.S. Gray NWHL 1/12/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J. E. Kelly 1/15/73
SIGNATURE AIR SYSCOM, BY DIRECTION DATE

ORIGINATOR Charles McBride 1/15/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING DOMESTIC UNIT LOADS OF MK 88 MOD 0 PROPELLANT GRAINS PALLETIZED IN ACCORDANCE WITH WR-53/845.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT VAN TRAILERS WITH THE TRAILERS TANDEM AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
3. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
4. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS AND ATTACH "EXPLOSIVES B" PLACARD TO BOTH SIDES, FRONT, AND BACK OF TRAILER.
5. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, COMMON CEMENT COATED OR CHEMICALLY ETCHED (ALTERNATE ANNULAR RING TYPE); STRAPPING, QQ-S-781, TYPE I, CLASS A.
6. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320.

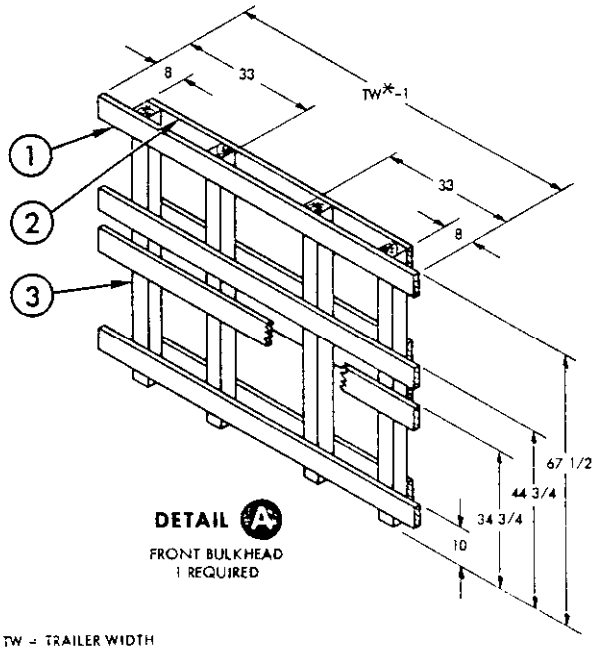
FTL 40 FT TRAILER (VAN)

1. TO PREVENT LONGITUDINAL MOVEMENT OF THE UNIT LOADS IN THE SECOND LAYER, THE TOP UNIT LOAD SHALL BE STRAPPED TO THE BOTTOM UNIT LOAD WHERE THE PATTERN CHANGES FROM TWO LAYERS HIGH TO ONE LAYER HIGH. THE 1 1/4 STRAPS, PIECE NO. 7, ENCIRCLE THE STACKED UNIT LOADS PASSING UNDER THE DECK OF THE BOTTOM UNIT LOAD AND OVER THE TOP OF THE UPPER UNIT LOAD. THE STRAPS ARE SEALED WITH TWO DOUBLE CRIMPED SEALS, PIECE NO. 8. THE TWO HIGH UNIT LOADS AT THE REAR OF THE TRAILER SHALL ALSO BE STRAPPED TOGETHER.
2. THE THICKNESS AND QUANTITY OF PIECES NO. 9 SHALL BE SUCH AS TO CAUSE THE REAR BLOCKING ASSEMBLY, DETAIL D, TO BEAR AGAINST THE UNIT LOADS AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.

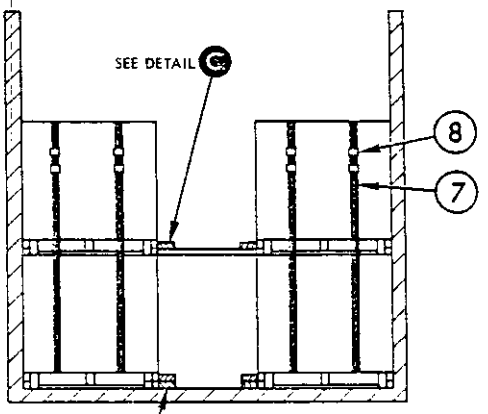
WARNING

DO NOT USE TRAILERS WITH ROLL UP REAR DOORS
WHEN BLOCKING BY THIS METHOD.

3. SWAY BRACE, DETAIL C, IS ASSEMBLED IN PLACE. POSITION SUPPORTS, PIECE NO. 5, IN UNIT LOAD FORK POCKETS AND NAIL STRINGERS, PIECE NO. 6, TO SUPPORTS WITH THREE 12d NAILS PER JOINT.



DETAIL A
FRONT BULKHEAD
1 REQUIRED

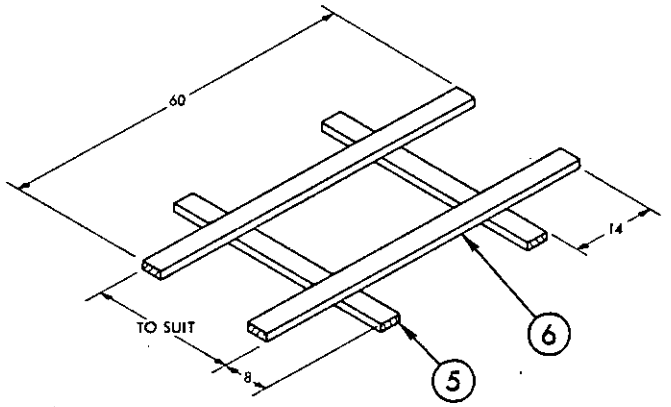


4 2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

SECTION B-B

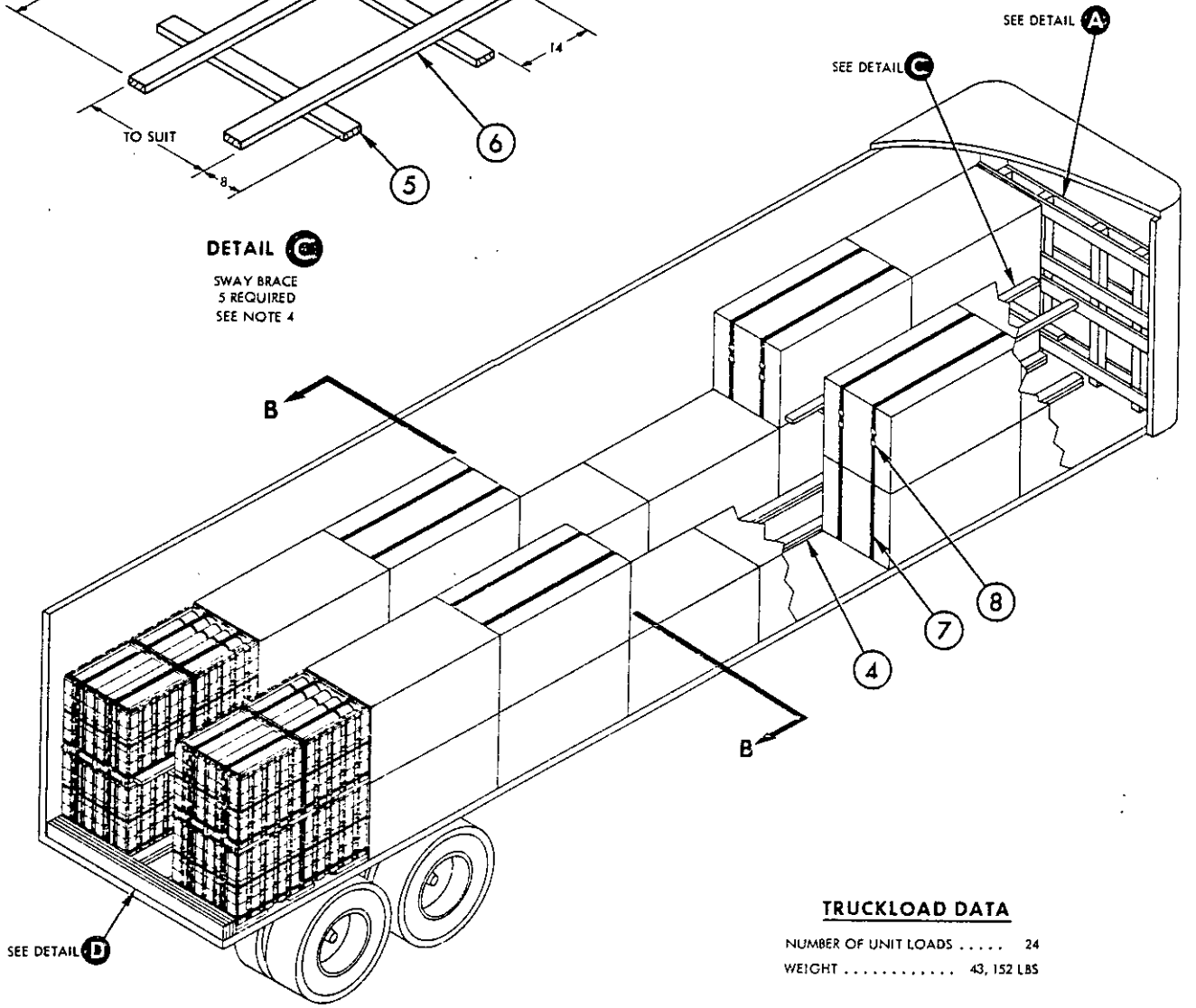
*TW = TRAILER WIDTH

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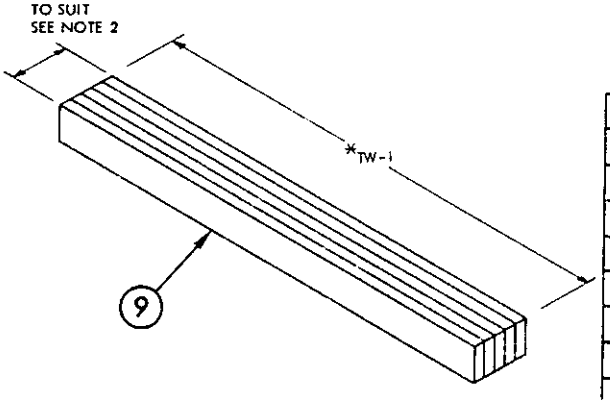
DETAIL C

SWAY BRACE
5 REQUIRED
SEE NOTE 4



TRUCKLOAD DATA

NUMBER OF UNIT LOADS 24
WEIGHT . . . . . 43, 152 LBS



DETAIL D

REAR BLOCKING ASSEMBLY
1 REQUIRED

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
9	CROSSMEMBER	THICKNESS TO SUIT 6 WIDE X *TW-1	AS REQD	LAMINATE	6/PIECE	TO SUIT
8	SEAL	FOR 1 1/4 STRAP	24	-	-	-
7	STRAP	1 1/4 X .035 X 20 FT	12	-	-	-
6	STRINGER	2 X 4 X 60	10	5	3/JOINT	12d
5	SUPPORT	2 X 4 X 40	10	SEE 6	-	-
4	SLEEPER	2 X 4 X 65	28	SEE SECTION B-B		
3	UPRIGHT	4 X 4 X 67 1/2	4	SEE 1 & 2	-	-
2	FWD HORIZONTAL	2 X 6 TO SUIT	4	3	3/JOINT	12d
1	AFT HORIZONTAL	2 X 6 X *TW-1	4	3	3/JOINT	12d

LIST OF MATERIALS AND NAILING DATA

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LTL 40 FT TRAILER (VAN)

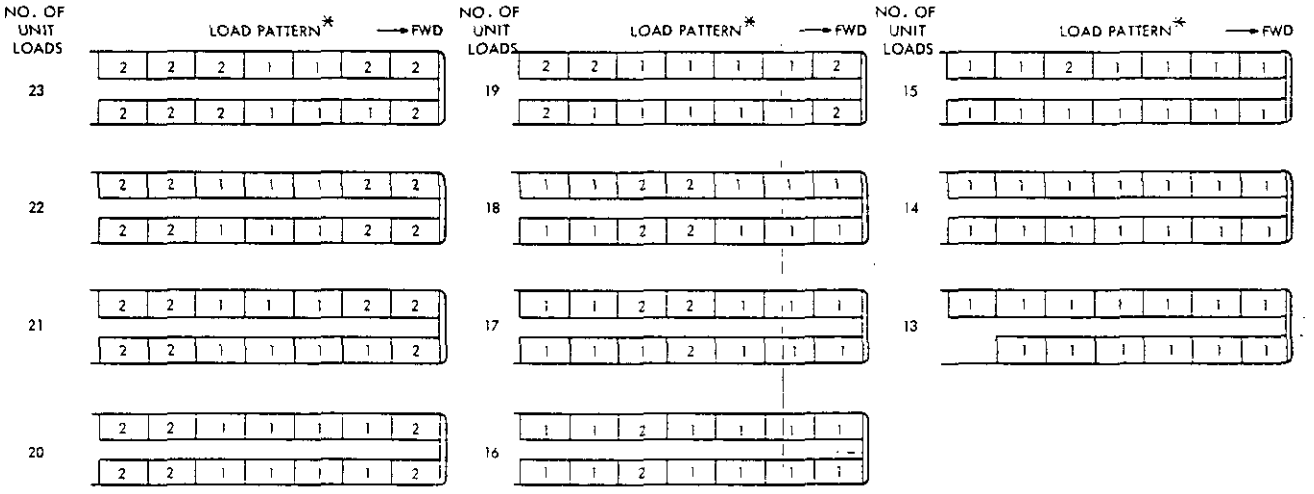
1. WHEN REQUIRED TO SHIP LESS THAN 24 UNIT LOADS, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF UNIT LOADS TO BE SHIPPED FROM TABLE I. LOAD THE TRAILER USING THIS PATTERN AND BLOCK AND BRACE THE LOAD USING THE PRINCIPLES OF THE LTL ON PAGE 5 AND THE FTL ON PAGE 3.
2. WHEN THE LOAD PATTERN REQUIRES UNIT LOADS TO BE SHIPPED TWO HIGH, CERTAIN UNIT LOADS SHALL BE STRAPPED TOGETHER. THIS IS TO ENSURE CONTINUOUS ENGAGEMENT OF THE STACKING FEATURES OF THE CRATES AND PREVENT LONGITUDINAL MOVEMENT IN THE SECOND LAYER. THE UNIT LOADS SHALL BE STRAPPED TOGETHER, AS SHOWN ON PAGE 3 IN THE FTL, WHERE THEY ARE TWO HIGH AT THE REAR OF THE TRAILER AND WHEN THE PATTERN CHANGES FROM TWO LAYERS HIGH TO ONE LAYER HIGH. THE 1 1/4 X .035 STRAPS ENCIRCLE THE STACKED UNIT LOADS PASSING UNDER THE DECK OF THE BOTTOM UNIT AND OVER THE TOP OF THE UPPER UNIT LOAD. THE STRAPS ARE SEALED WITH TWO DOUBLE-CRIMPED SEALS. SEE FTL ON PAGE 3 FOR STRAPPED TWO HIGH UNIT LOADS.
3. THE THICKNESS AND QUANTITY OF PIECES NO. 7 SHALL BE SUCH AS TO CAUSE THE REAR BLOCKING ASSEMBLY, DETAIL H, TO BEAR AGAINST THE UNIT LOAD AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.

WARNING

DO NOT USE TRAILERS WITH ROLL-UP DOORS
WHEN BLOCKING BY THIS METHOD.

4. THE LTL ON PAGE 5 IS FOR 13 UNIT LOADS. THIS DOES NOT LIMIT LTL SHIPMENTS TO THIS QUANTITY. ACTUAL SHIPPING REQUIREMENTS WILL DETERMINE THE NUMBER OF UNIT LOADS TO BE SHIPPED. THE LOAD PATTERN AND DUNNAGING SHALL BE IN ACCORDANCE WITH THE APPROPRIATE PARTS OF THIS MILITARY STANDARD AND THE GENERAL DOCUMENT.

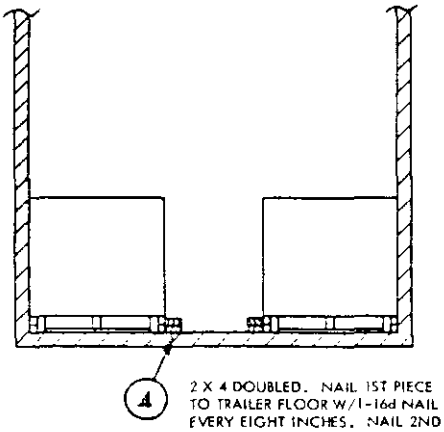
TABLE I



12 OR LESS:

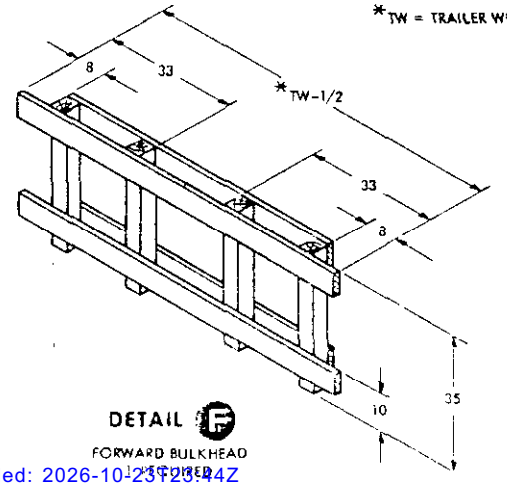
POSITION UNIT LOADS IN TRAILER USING A "DOUBLE ROW" PATTERN. TO PREVENT REARWARD MOVEMENT, INSTALL REAR BLOCKING, DETAIL G, AT THE END OF EACH ROW.

*LOAD PATTERN SHOWS FLOOR PLAN FOR TRAILER. NUMBER IN BLOCKS INDICATES NUMBER OF UNIT LOADS IN LAYER.



A — 2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY EIGHT INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

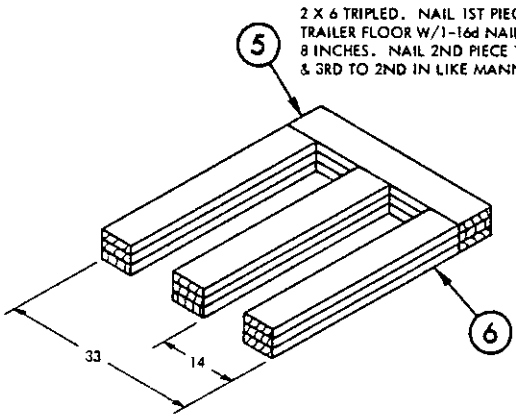
SECTION C-C



DETAIL F

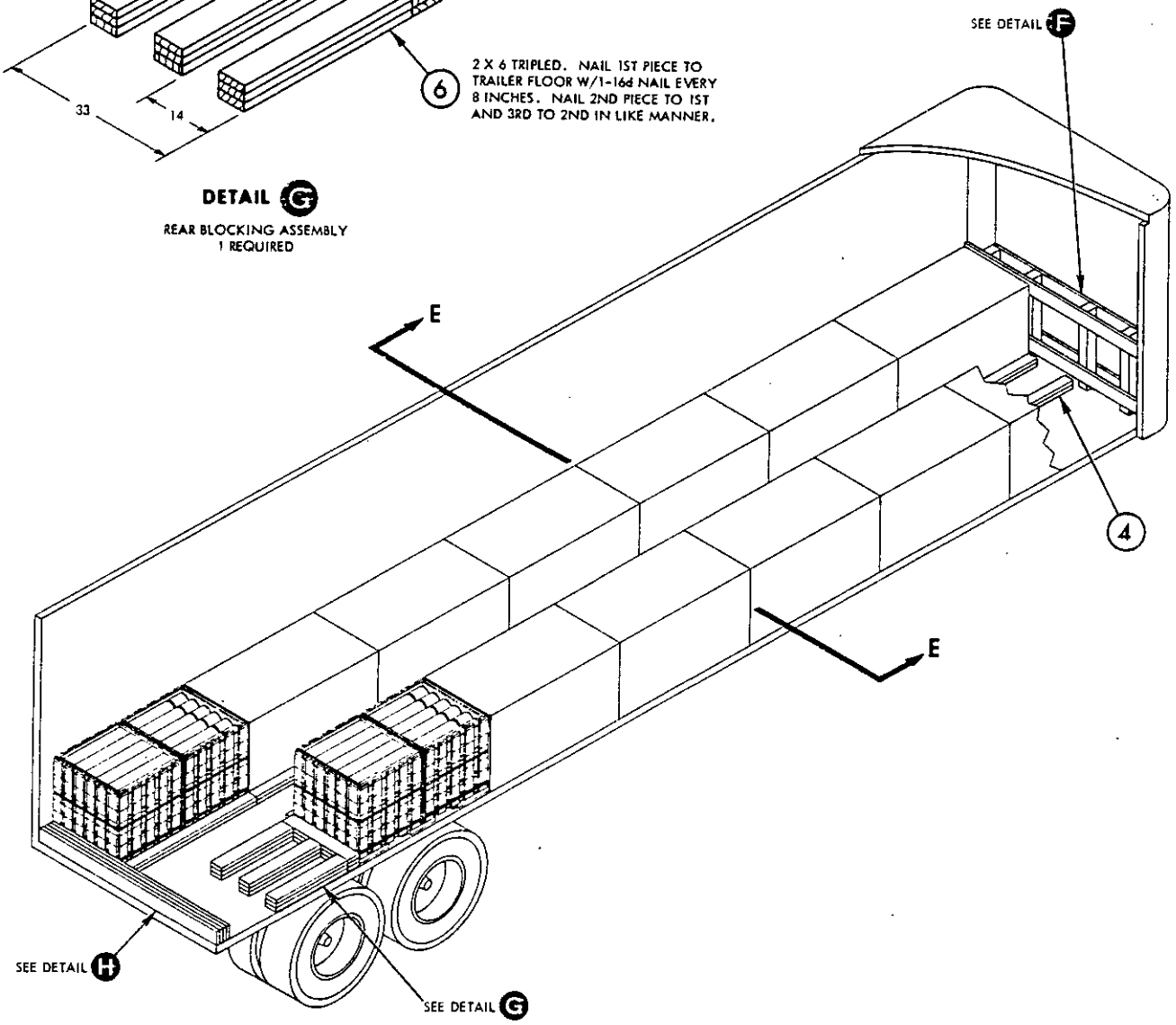
FORWARD BULKHEAD

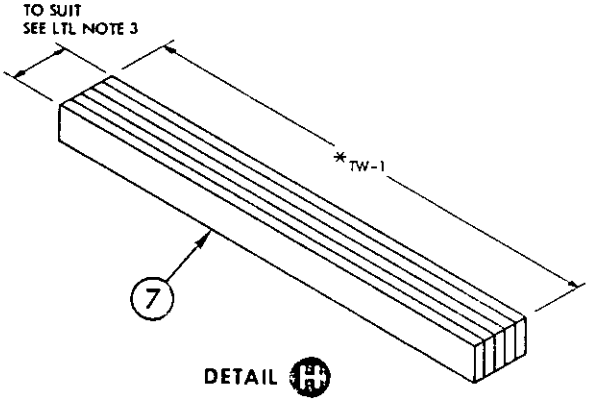
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DETAIL G

REAR BLOCKING ASSEMBLY
1 REQUIRED





DETAIL H

REAR BLOCKING ASSEMBLY
1 REQUIRED

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
7	CROSSMEMBER	THICKNESS TO SUIT 6 WIDE X TW*-1	AS REQD	LAMINATE	6/PIECE	TO SUIT
6	BACKUP CLEAT	2 X 6 X 36	9	SEE DETAIL G		
5	CROSSMEMBER	2 X 6 X 33	3	SEE DETAIL G		
4	SLEEPER	2 X 4 X 65	26	SEE DETAIL E-E		
3	VERTICAL	4 X 4 X 35	4	SEE 1 & 2	-	-
2	FWD HORIZONTAL	2 X 6 TO SUIT	2	3	3/JOINT	12d
1	AFT HORIZONTAL	2 X 6 X TW*-1	2	3	3/JOINT	12d

LIST OF MATERIALS AND NAILING DATA

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